Gépi látás és képfeldolgozás az egészségügyben féléves feladat

Készítette: Sándor Zsombor

Használt modellek

- ML modellek:
  - -KNN
  - -SVM
- Dl modellek:
  - -MobilnetV2
  - -LeNet
  - -Densenet121

Előfeldolgozás:

Újra méretezés

Szürke árnyalatossá alakítás

Ml modellek számára lapítás

Normalizálás [0-1] tartományra

Hisztogram kiegyenlítés

Augmentáció:

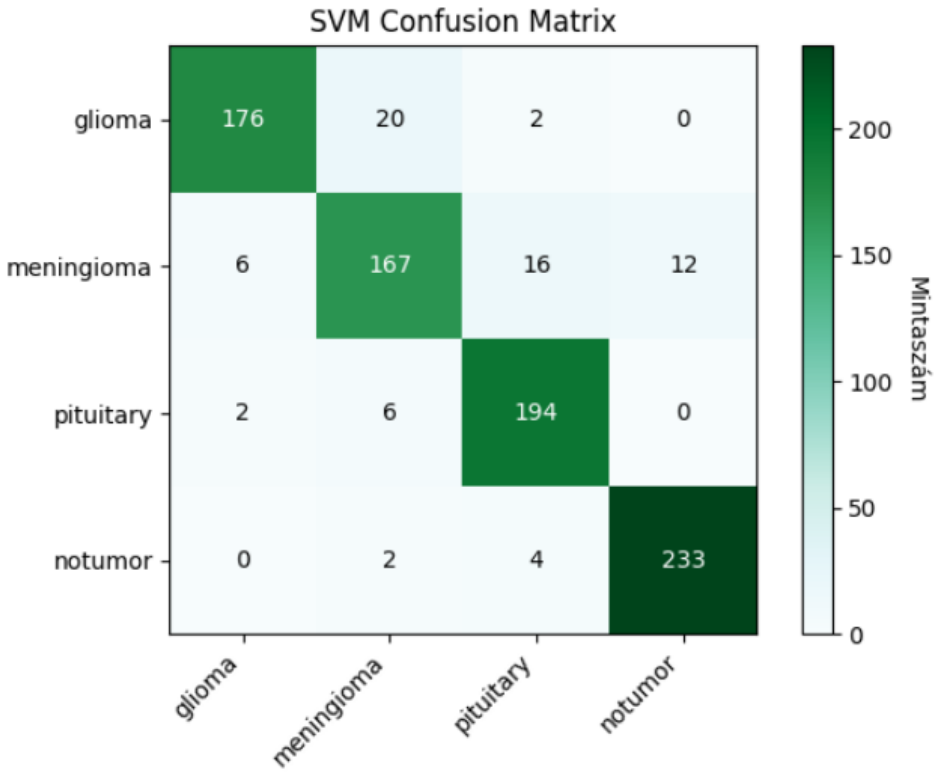
Keras – ImageDataGenerator:

- véletlenszerű vízszintes tükrözés
- +-15% kicsinyítés/nagyítás
- vízszintes/függőleges eltolás 10%ig

```
datagen = ImageDataGenerator(

    horizontal_flip=True,
    zoom_range=0.15,
    width_shift_range=0.1,
    height_shift_range=0.1
)
```

SVM

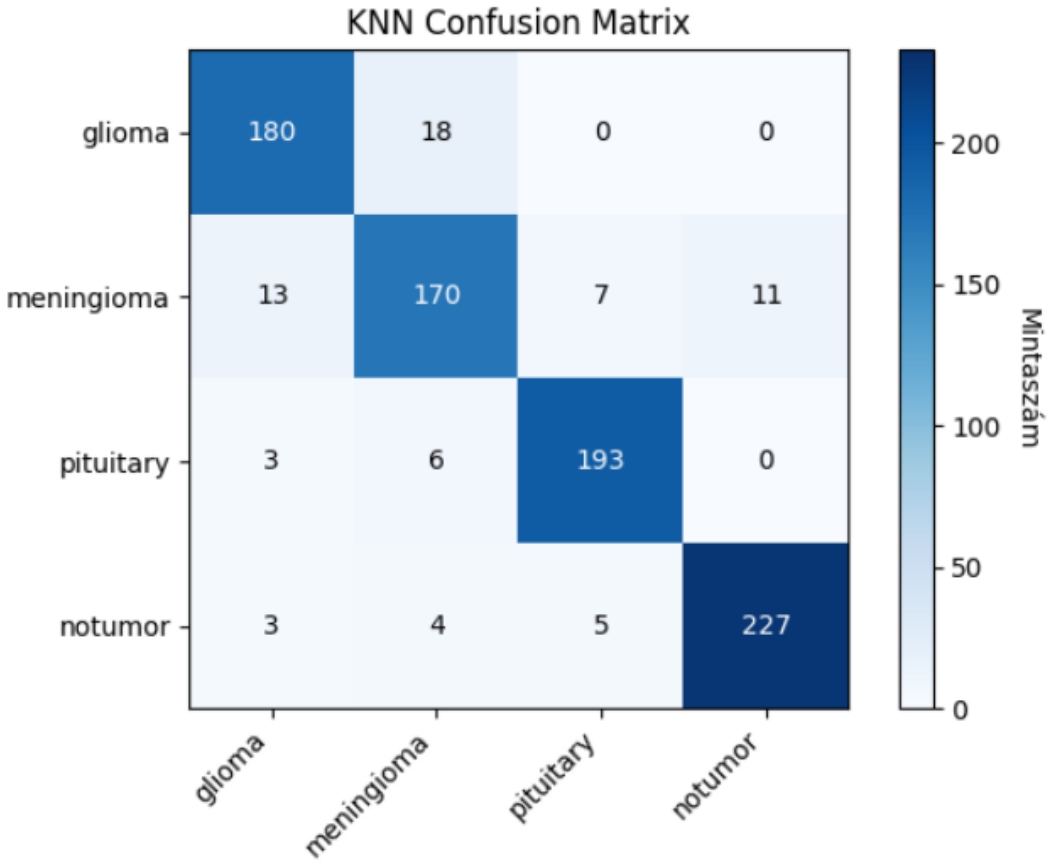



```
>> Training SVM...
SVM — Test Accuracy: 0.9167
              precision    recall  f1-score   support

      glioma       0.96      0.89      0.92       198
  meningioma       0.86      0.83      0.84       201
   pituitary       0.90      0.96      0.93       202
     notumor       0.95      0.97      0.96       239

    accuracy                           0.92       840
   macro avg       0.92      0.91      0.91       840
weighted avg       0.92      0.92      0.92       840
```

K-nearest neighbors



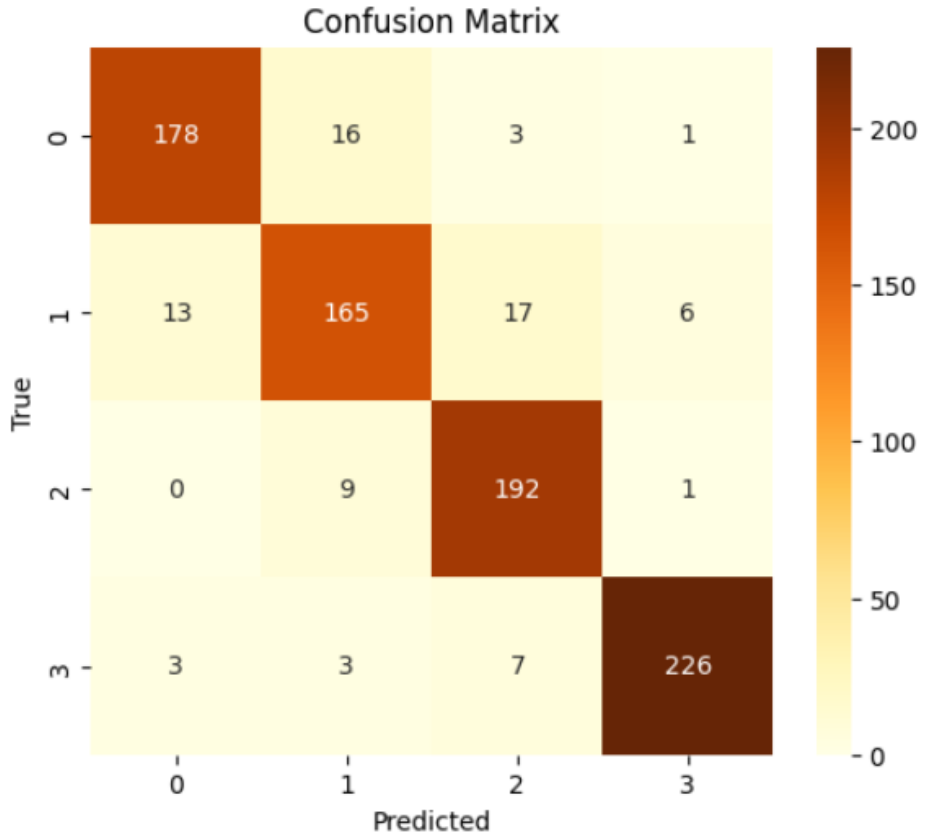

```
>> Training KNN...
KNN — Test Accuracy: 0.9167
              precision    recall  f1-score   support

      glioma       0.90      0.91      0.91       198
  meningioma       0.86      0.85      0.85       201
   pituitary       0.94      0.96      0.95       202
     notumor       0.95      0.95      0.95       239

    accuracy                           0.92       840
   macro avg       0.91      0.92      0.91       840
weighted avg       0.92      0.92      0.92       840
```

MobileNetV2

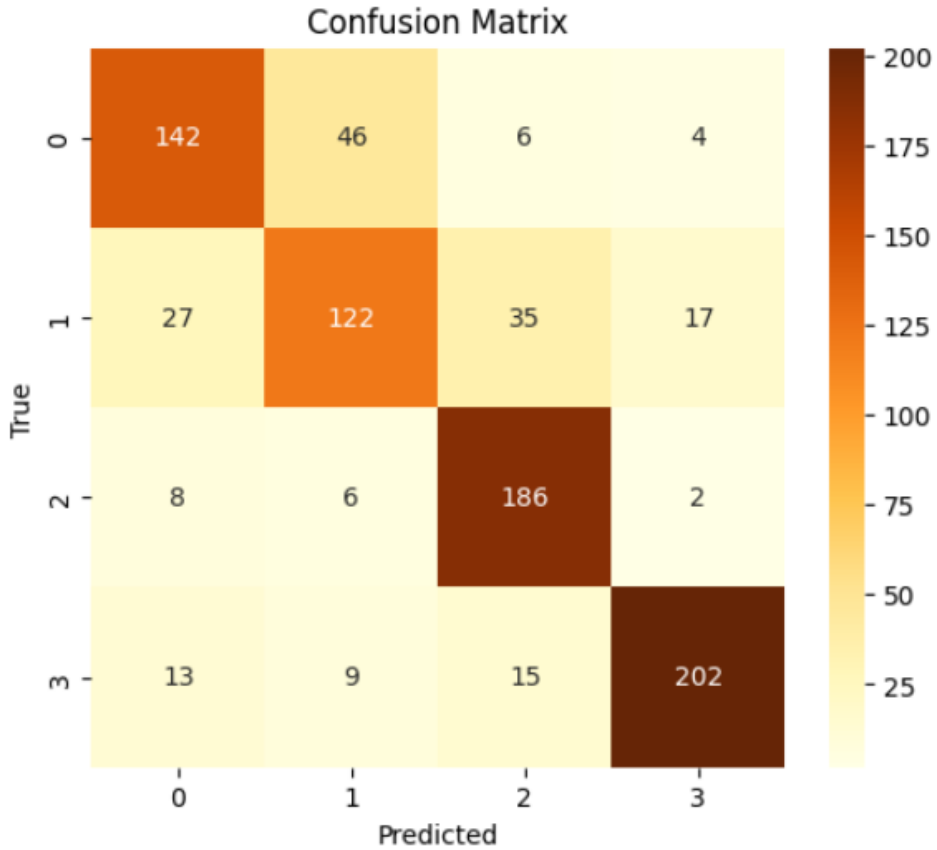
	precision	recall	f1-score	support
glioma	0.92	0.90	0.91	198
meningioma	0.85	0.82	0.84	201
pituitary	0.88	0.95	0.91	202
notumor	0.97	0.95	0.96	239
accuracy			0.91	840
macro avg	0.90	0.90	0.90	840
weighted avg	0.91	0.91	0.91	840





LeNet

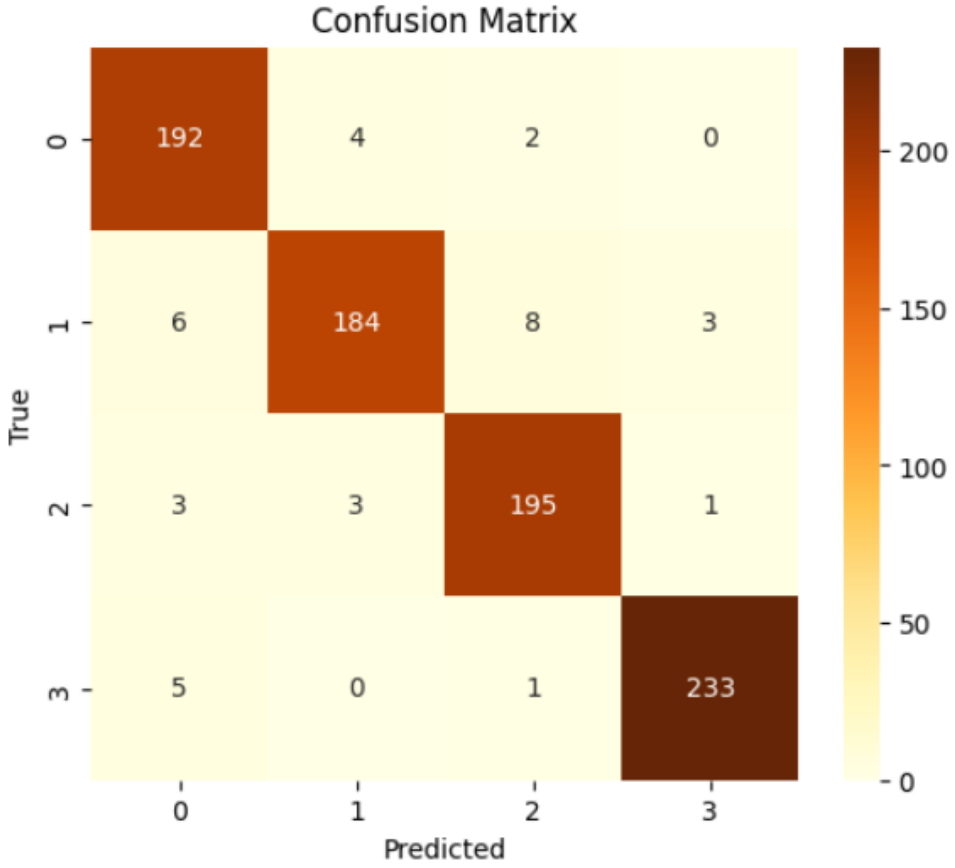
	precision	recall	f1-score	support
glioma	0.75	0.72	0.73	198
meningioma	0.67	0.61	0.64	201
pituitary	0.77	0.92	0.84	202
notumor	0.90	0.85	0.87	239
accuracy			0.78	840
macro avg	0.77	0.77	0.77	840
weighted avg	0.78	0.78	0.77	840

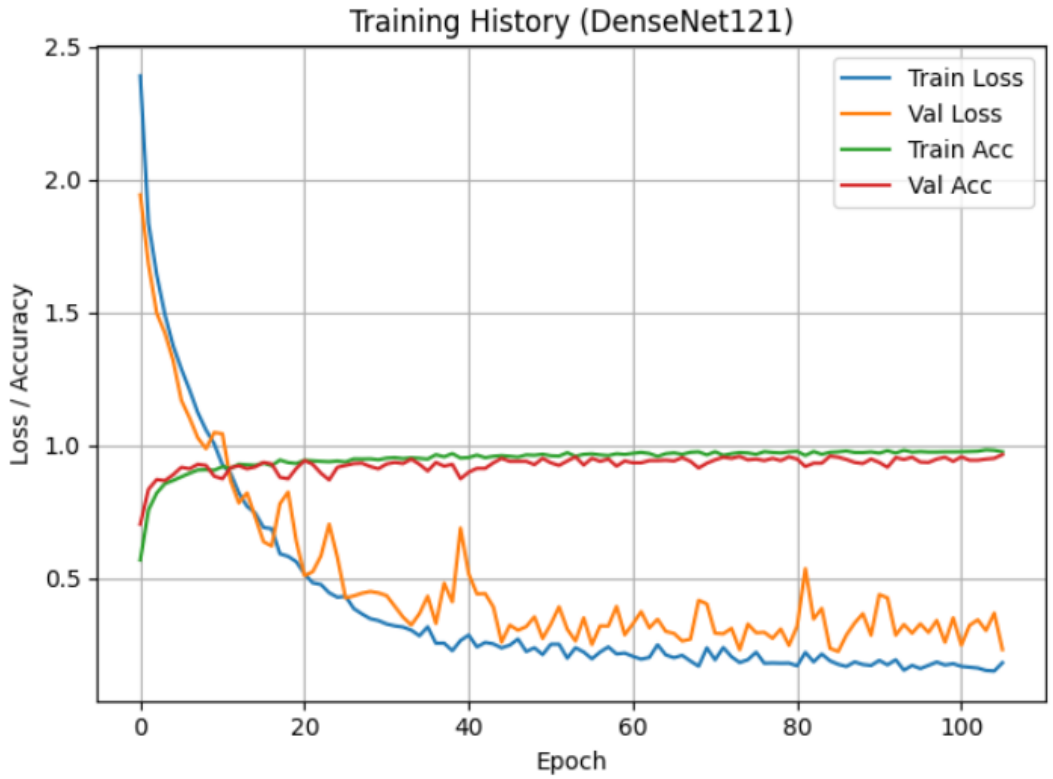




Densenet121

	precision	recall	f1-score	support
glioma	0.93	0.97	0.95	198
meningioma	0.96	0.92	0.94	201
pituitary	0.95	0.97	0.96	202
notumor	0.98	0.97	0.98	239
accuracy			0.96	840
macro avg	0.96	0.96	0.96	840
weighted avg	0.96	0.96	0.96	840





Köszönöm a figyelmet!